PLANETARY STABILITY AND HABITABILITY IN TRIPLE STAR SYSTEMS

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OVERVIEW

Dynamical Model

Numerical Framework

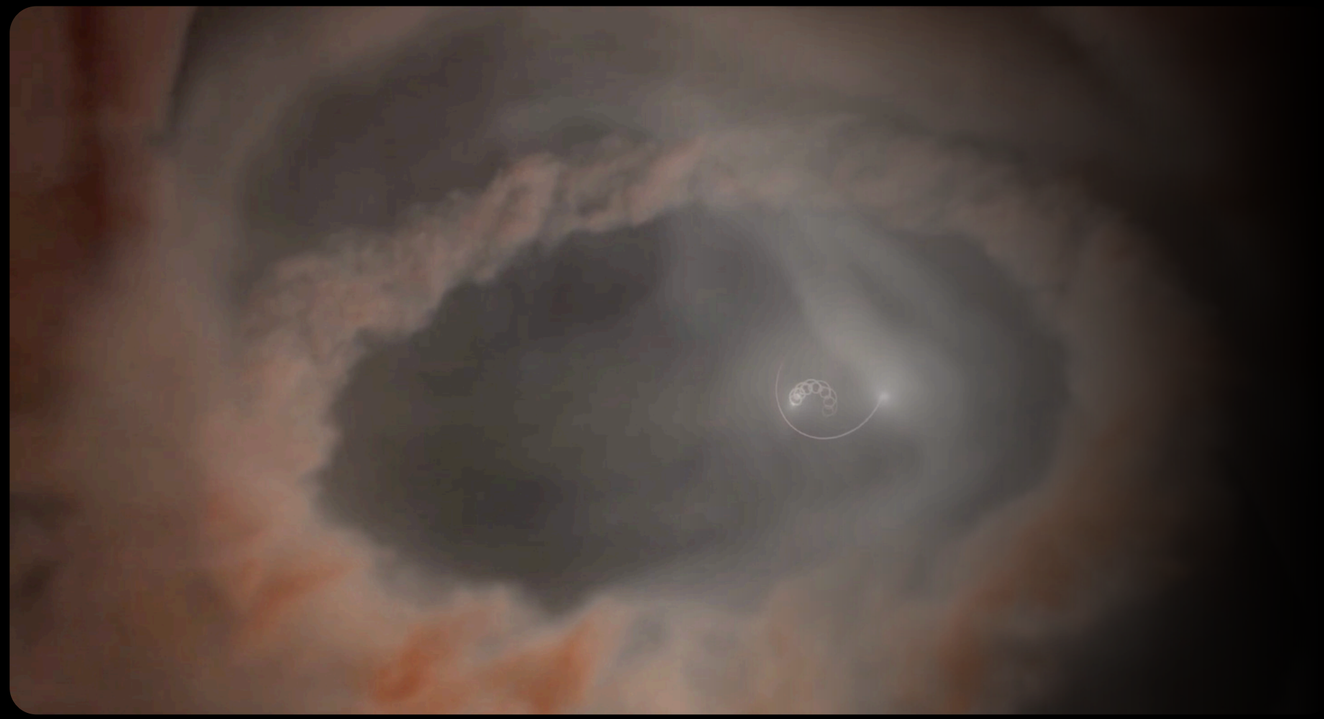
Radiative Model

Results

Discussion



Kepler-186f, the First Earth-size Planet in the Habitable Zone (JPL NASA)



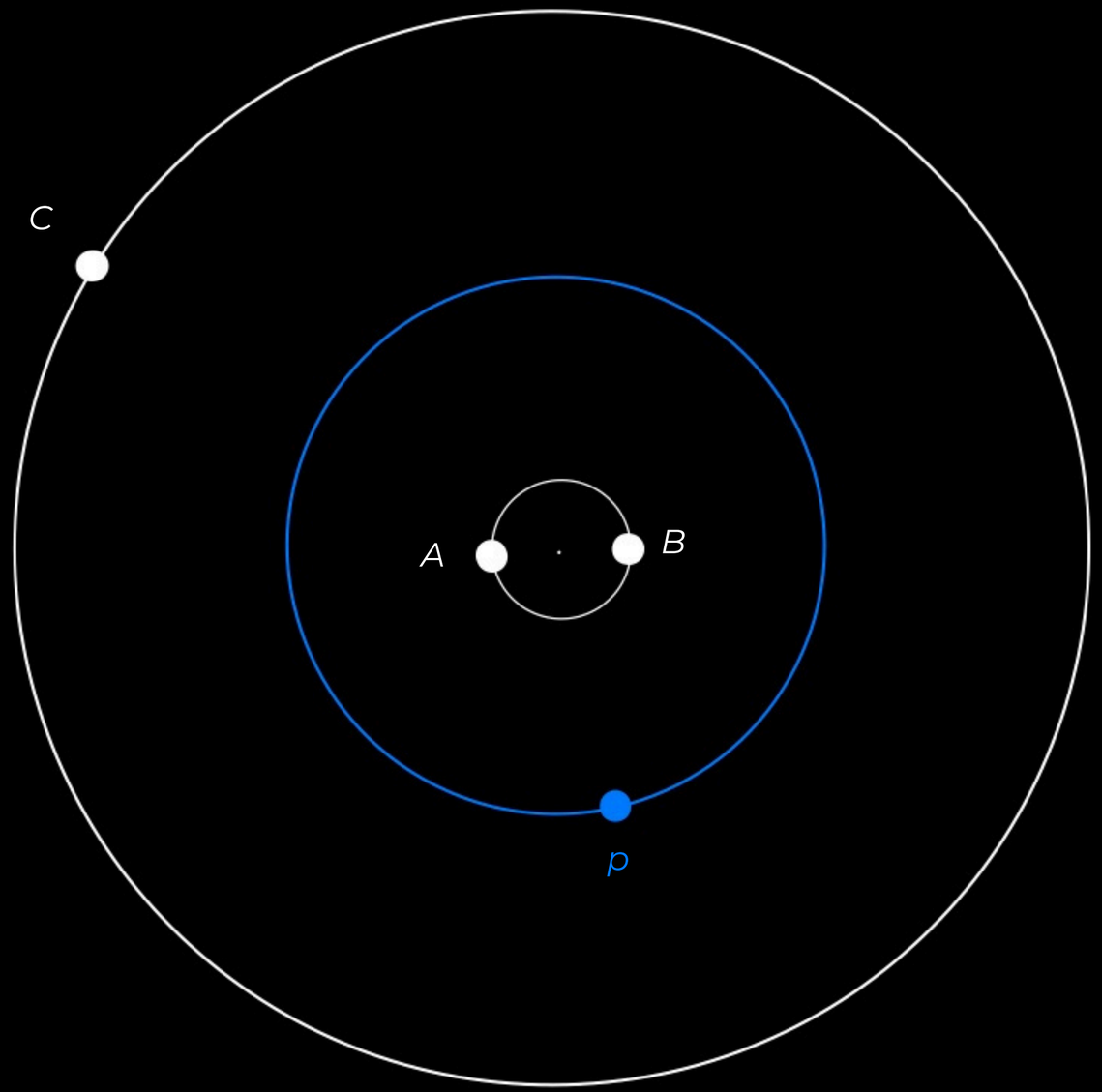
GW Orionis (triple stellar system), ESO

INTRODUCTION

Understand the influence of the companion star on planetary habitability in hierarchical triple star systems

→ **Stability condition:** long-term orbital survival of the planet

→ **Radiative condition:** planet surface temperature compatible with liquid water:



DYNAMICAL CONFIGURATION

Hierarchical triple system

- Inner binary: Star A & star B
- External companion: Star C

Planet p

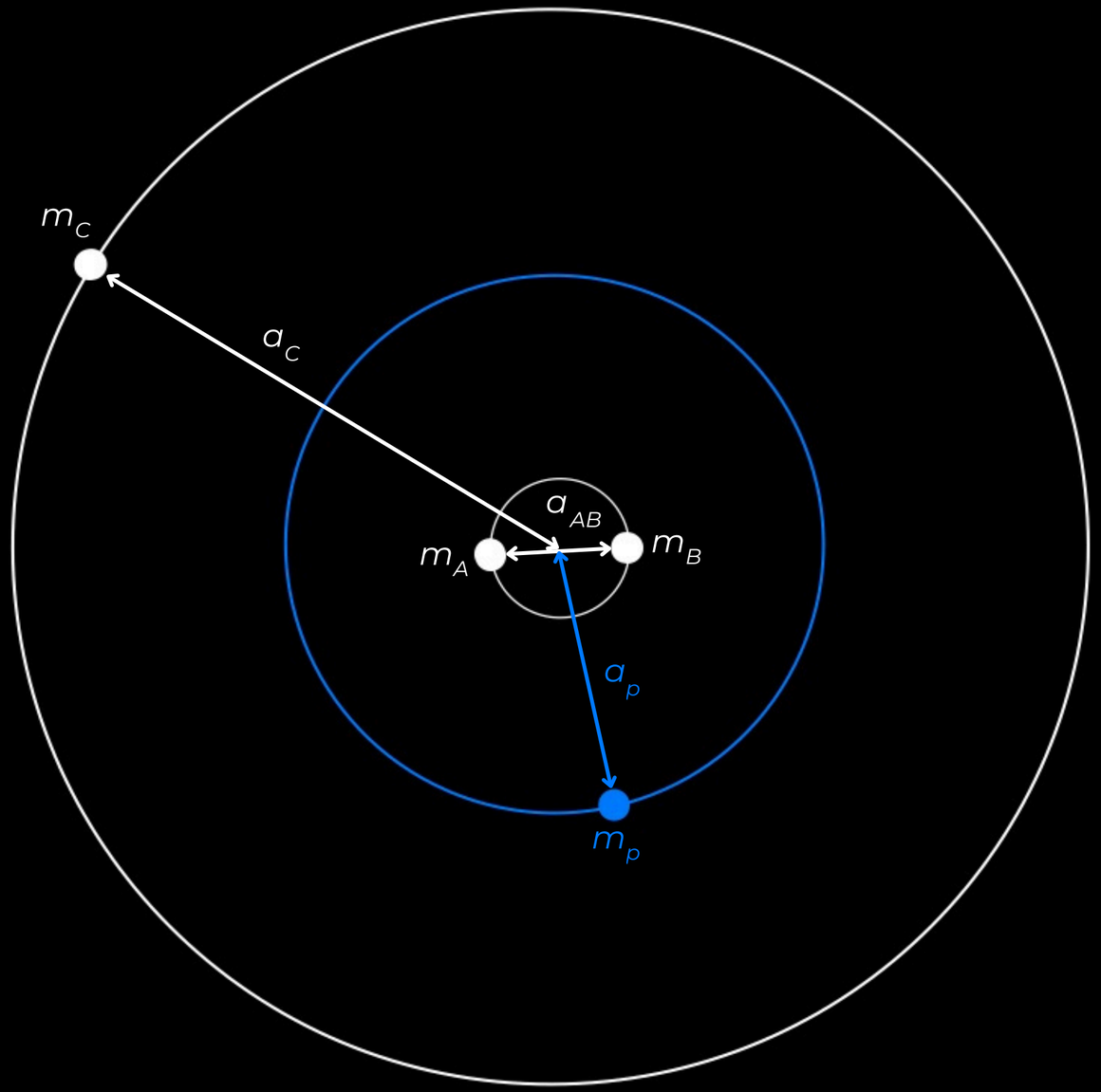
- Circumbinary, or “type P”
- Orbits the barycentre of the inner binary

—

Dynamical constraints (no ejection or collision)
→ defines the **Stability Zone (SZ)**

Radiative constraints (liquid water at the surface)
→ defines the **Habitable Zone (HZ)** Kasting et al. (1993)

Habitability is the overlap between the SZ and the HZ



FIXED PARAMETERS

Binary AB

$m_A = 2M_\odot$ $\quad a_{AB} = 0.1$ AU

$m_B = 1M_\odot$ $\quad e_{AB} = 0$

Companion C

$a_C = 15$ AU

Planet p

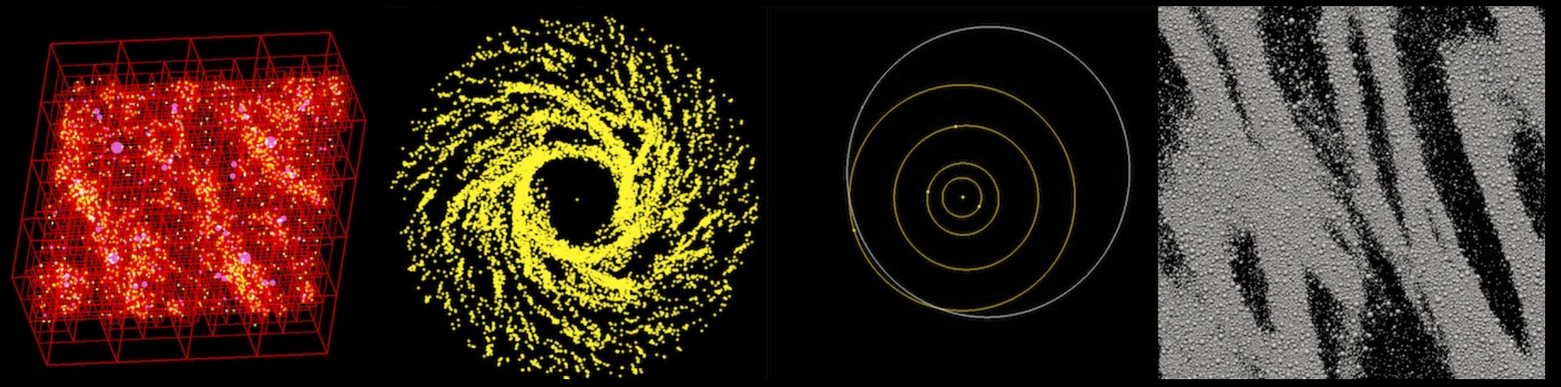
$m_p = 1M_\oplus$ $\quad e_p = 0$

VARIABLE PARAMETERS

Companion C m_C from 0.1 to 2.3 $M_\odot$

e_C from 0 to 0.6

INTEGRATOR



REBOUND

Python program using **REBOUND package** Rein & Liu (2012)

WHFast Symplectic integrator, allowing us to study the dynamics over long timescales Rein & Tamayo (2015)

Timestep $dt = \frac{P_{AB}}{20}$ Duncan et al. (1998)

Integration time $n = 10^4 \,\mathrm{periods}$ Holman & Wiegert (1999)

MAPPING THE STABILITY ZONE

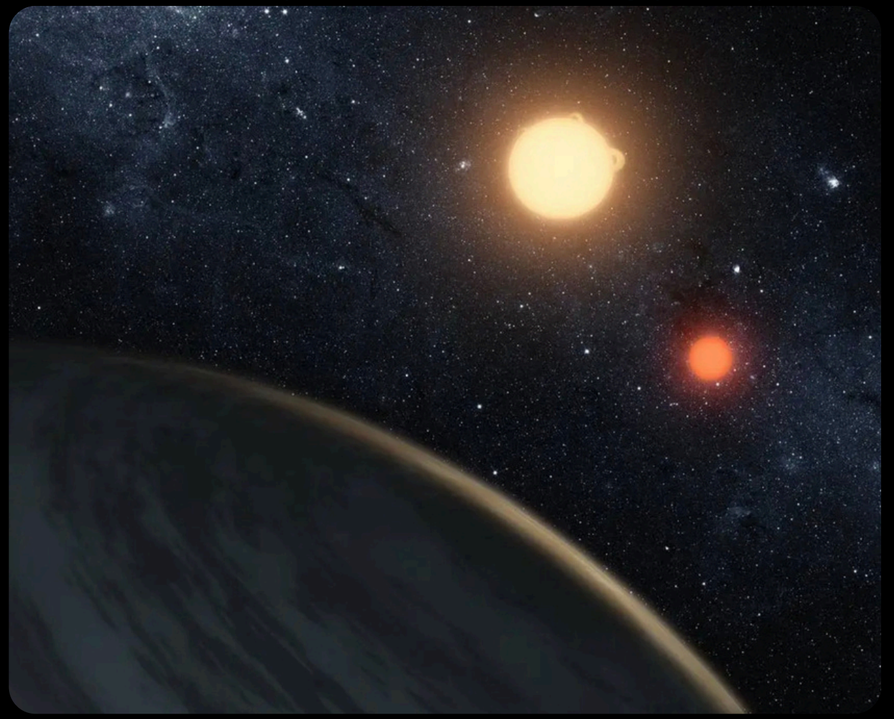
Spatial Discretization

- 1D grid between the inner binary AB and companion C
- Space divided into *k* sampled planetary semi-major axis (testing different a_p)

Robustness

- *N* simulations per tested distance
- Randomized initial phases (true anomaly, arguments) to prevent bias from specific initial conditions

Kepler-16b (circumbinary planet), NASA



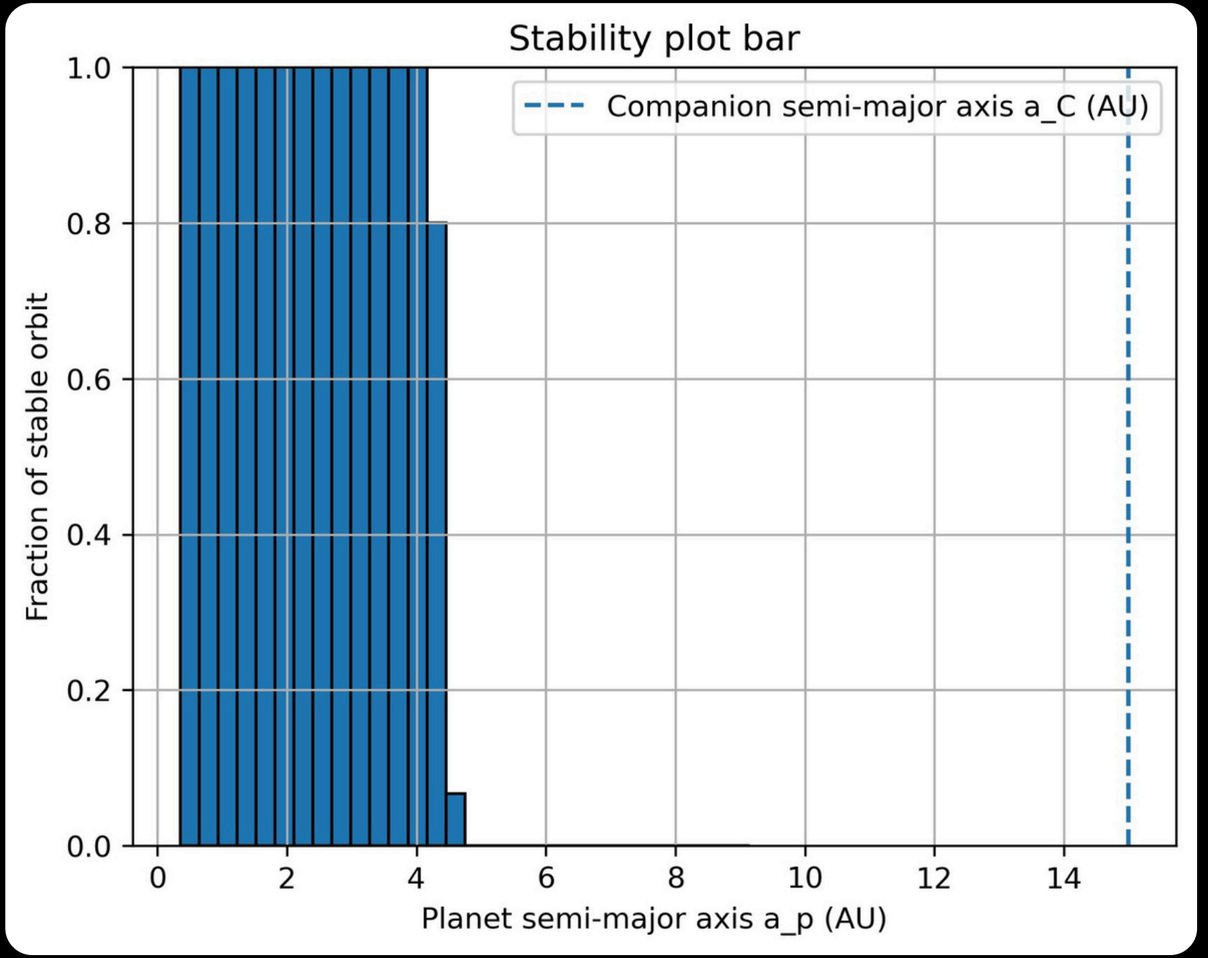
DEFINING & MEASURING STABILITY

Instability criteria

- The semi-major axis a_p becomes too large
- The eccentricity reaches $e_p \geq 1$ *(hyperbolic ejection)*

Stability fraction

- $\text{Ratio} = \dfrac{\text{Number of stable orbits}}{\text{Total tested orbits}}$
- The planet is considered stable if the ratio is *> 0.9*



RADIATIVE MODEL

Assumptions

- ➜ Radiative equilibrium
- ➜ Black body emission
- ➜ Wavelength independent
- ➜ No temperature diffusion
- ➜ No eclipsing
- ➜ Disk planetary cross section
- ➜ Constant star luminosity
- ➜ Earth like atmosphere

Mass-Luminosity relation

Duric 2004

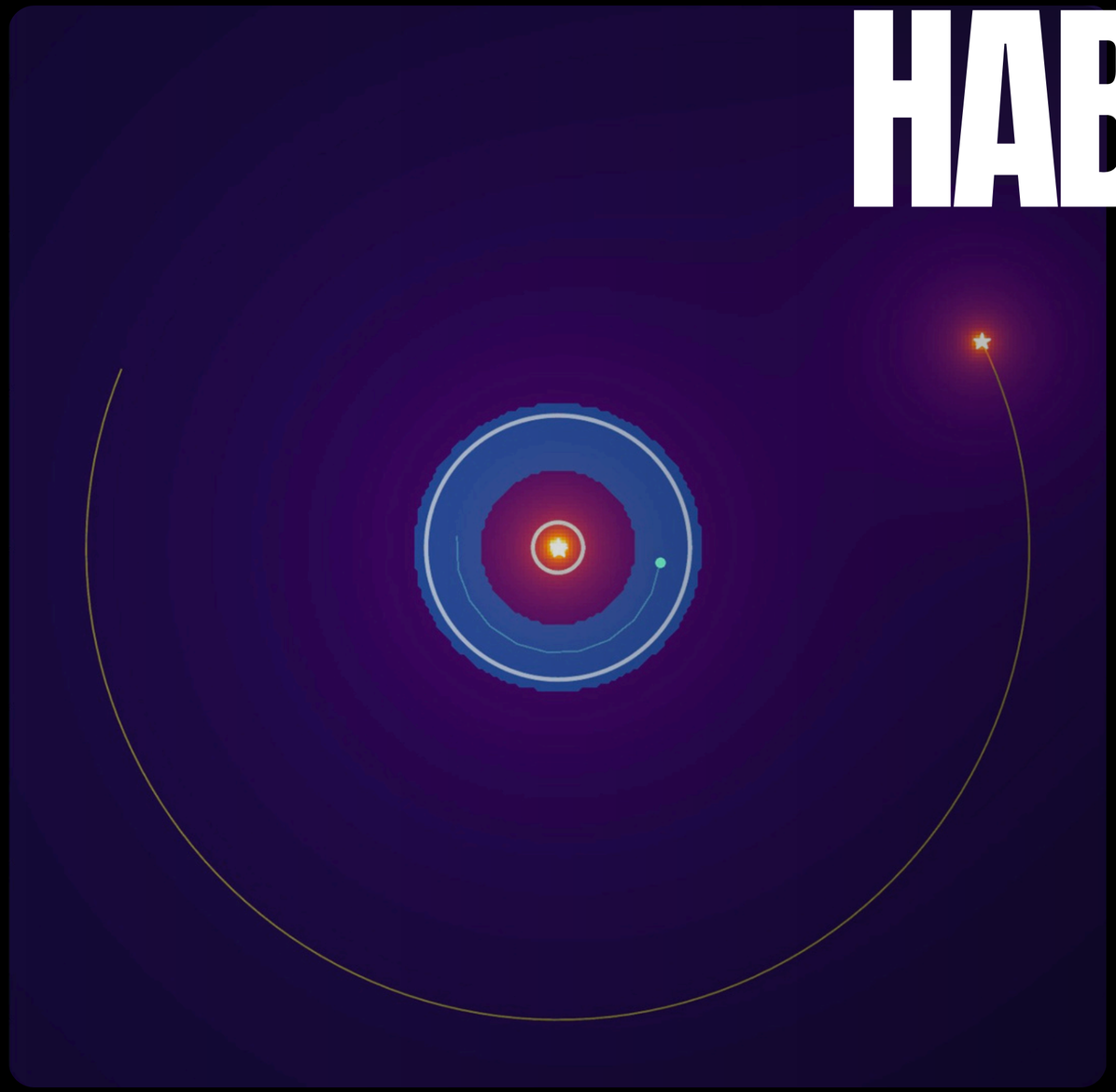
$$0.23\,L_\odot\left(\frac{M_i}{M_\odot}\right)^{2.3} \quad \text{if } M_i < 0.43\,M_\odot$$

$$L_\odot\left(\frac{M_i}{M_\odot}\right)^{4} \quad \text{if } M_i \geq 0.43\,M_\odot$$

Equilibrium temperature

$$T_{eq} = f_{gh}\left[\frac{1-\alpha}{16\pi\sigma}\left(\sum_i \frac{L_i}{d_i^2}\right)\right]^{1/4}$$

$$\alpha_\oplus \simeq 0.30 \quad f_{\mathrm{gh},\oplus} = \frac{T_{\mathrm{surf},\oplus}}{T_{\mathrm{eff},\oplus}} = \frac{288\,\mathrm{K}}{255\,\mathrm{K}} \simeq 1.13$$

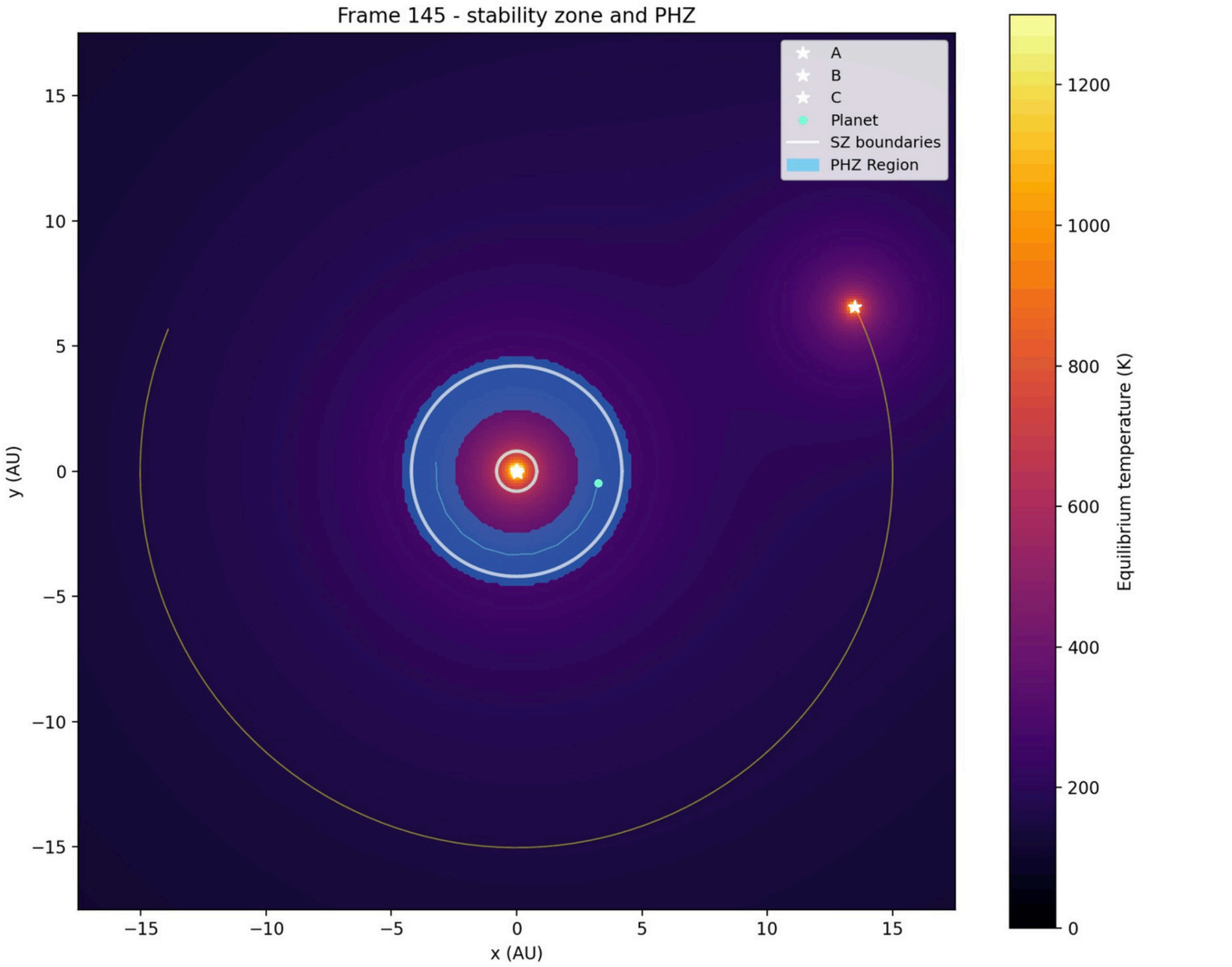


HABITABLE ZONE

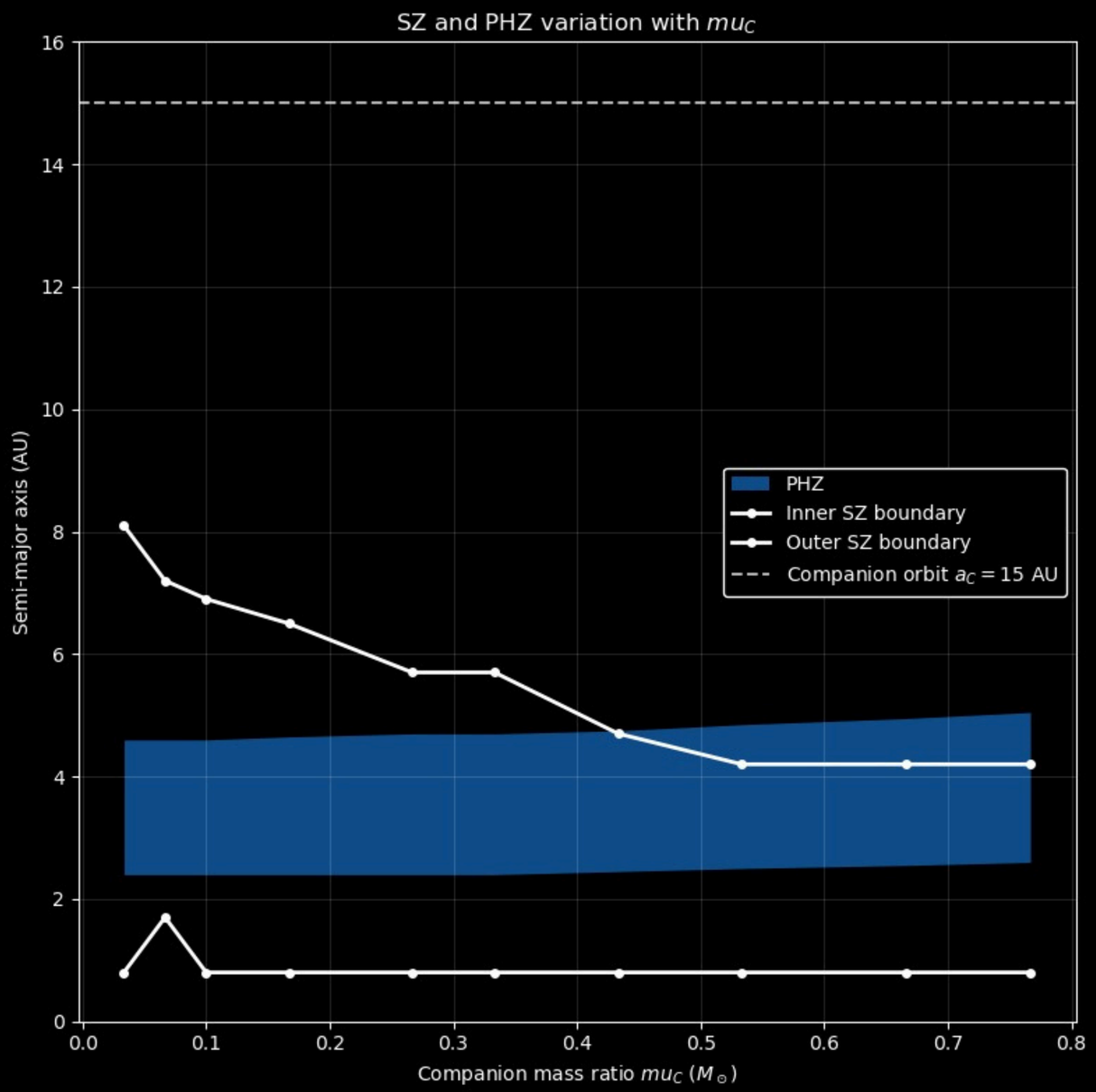
Range of orbital distances within which liquid water can exist on the surface of a rocky planet Kasting et al. 1993

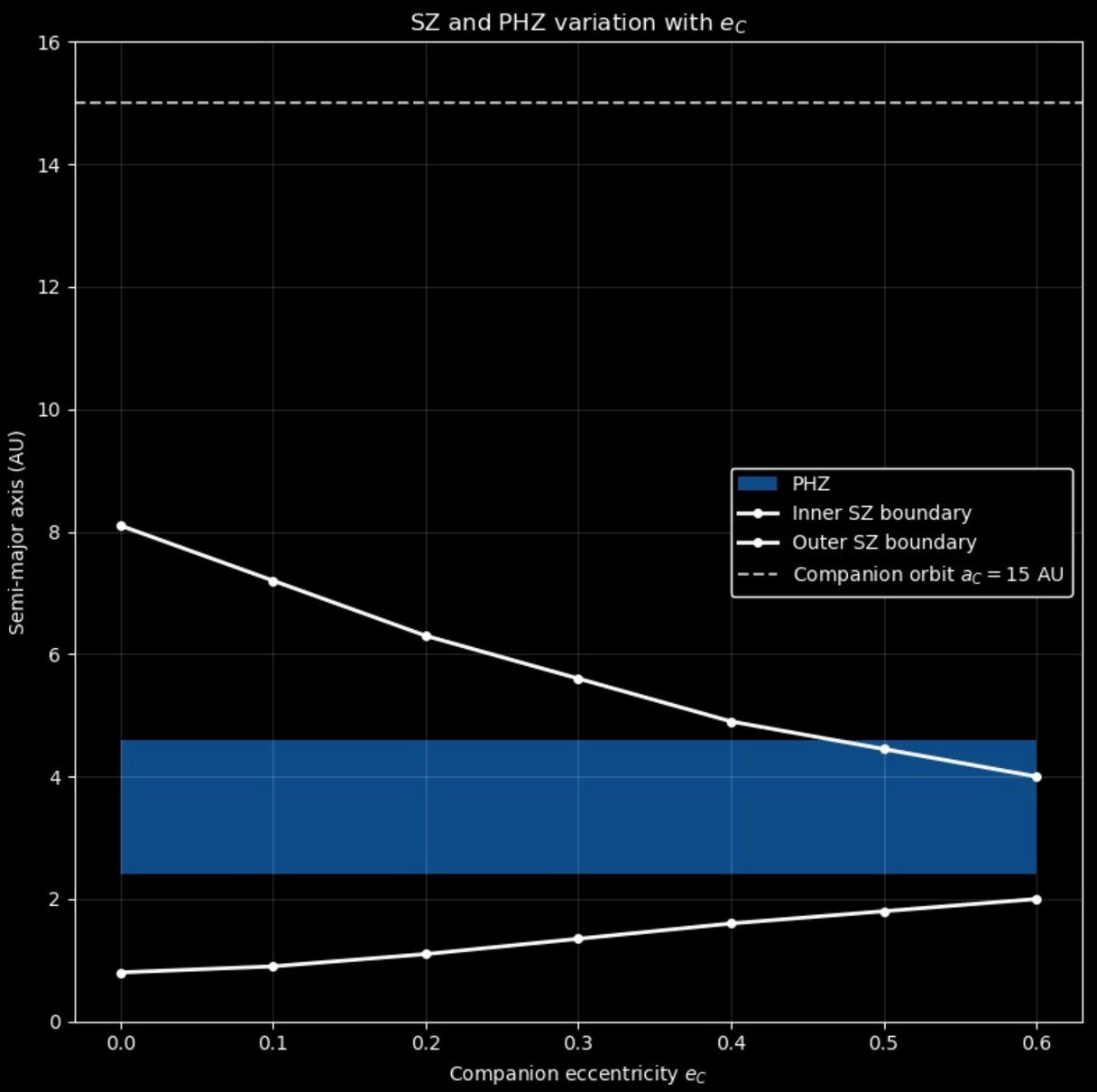
Permanent Habitable Zone (PHZ) Region where a planet remains within the habitable-zone temperature limits during its entire orbit, despite orbital and flux variations Eggl et al. 2012

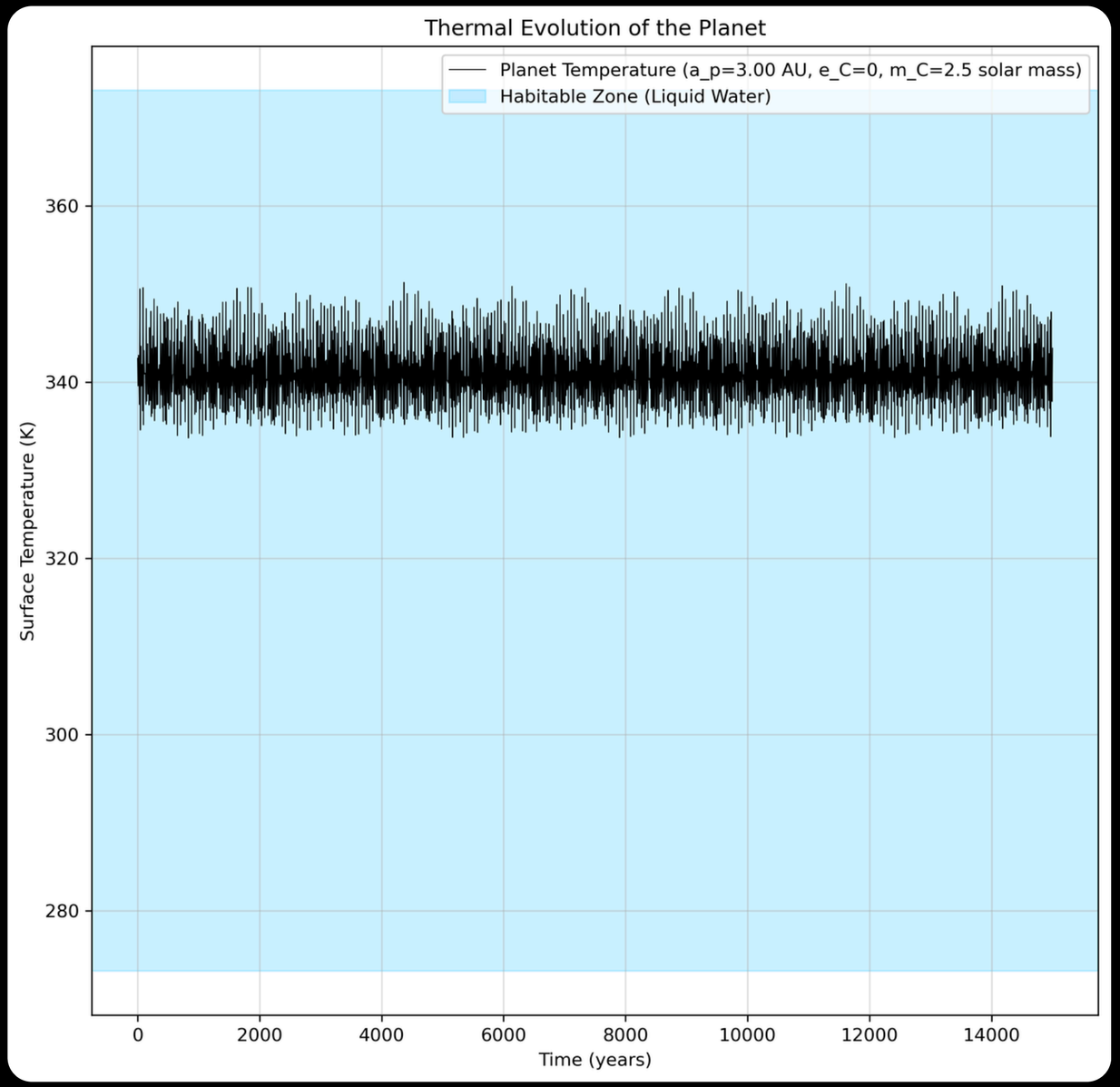
RESULTS

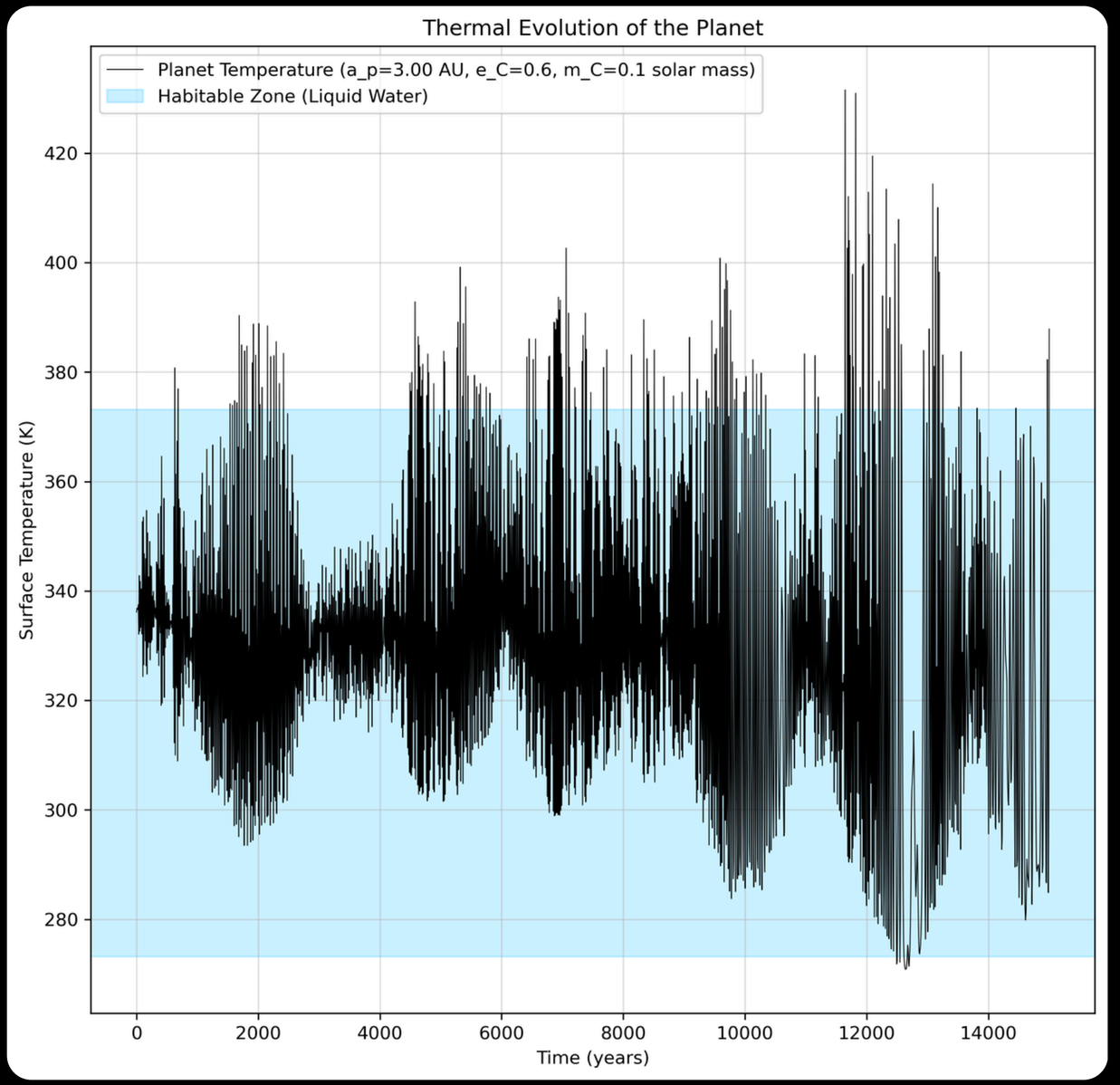


Temperature map for e_C = 0 and m_C = 1.5 solar mass


SZ and PHZ variation with mu_C
Semi-major axis (AU)
Companion mass ratio mu_C ($M_\odot$)
PHZ
Inner SZ boundary
Outer SZ boundary
Companion orbit $a_C = 15$ AU
0
2
4
6
8
10
12
14
16
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


SZ and PHZ variation with e_C
Semi-major axis (AU)
Companion eccentricity e_C
PHZ
Inner SZ boundary
Outer SZ boundary
Companion orbit $a_C = 15$ AU
0
2
4
6
8
10
12
14
16
0.0
0.1
0.2
0.3
0.4
0.5
0.6


Thermal Evolution of the Planet
Planet Temperature (a_p=3.00 AU, e_C=0, m_C=2.5 solar mass)
Habitable Zone (Liquid Water)
Surface Temperature (K)
Time (years)
280
300
320
340
360
0
2000
4000
6000
8000
10000
12000
14000


Thermal Evolution of the Planet
Planet Temperature (a_p=3.00 AU, e_C=0.6, m_C=0.1 solar mass)
Habitable Zone (Liquid Water)
Surface Temperature (K)
Time (years)
280
300
320
340
360
380
400
420
0
2000
4000
6000
8000
10000
12000
14000

CONCLUSION AND DISCUSSION

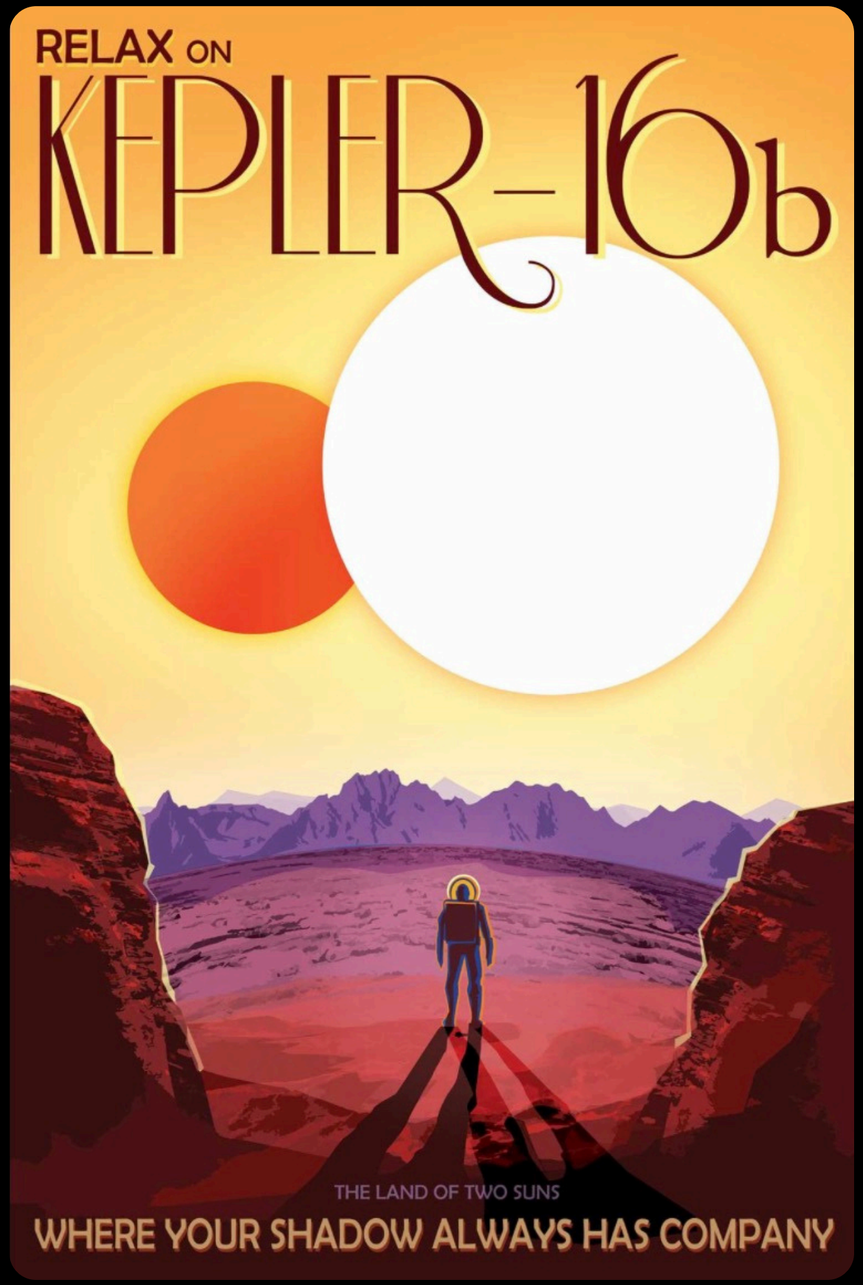
Limitations

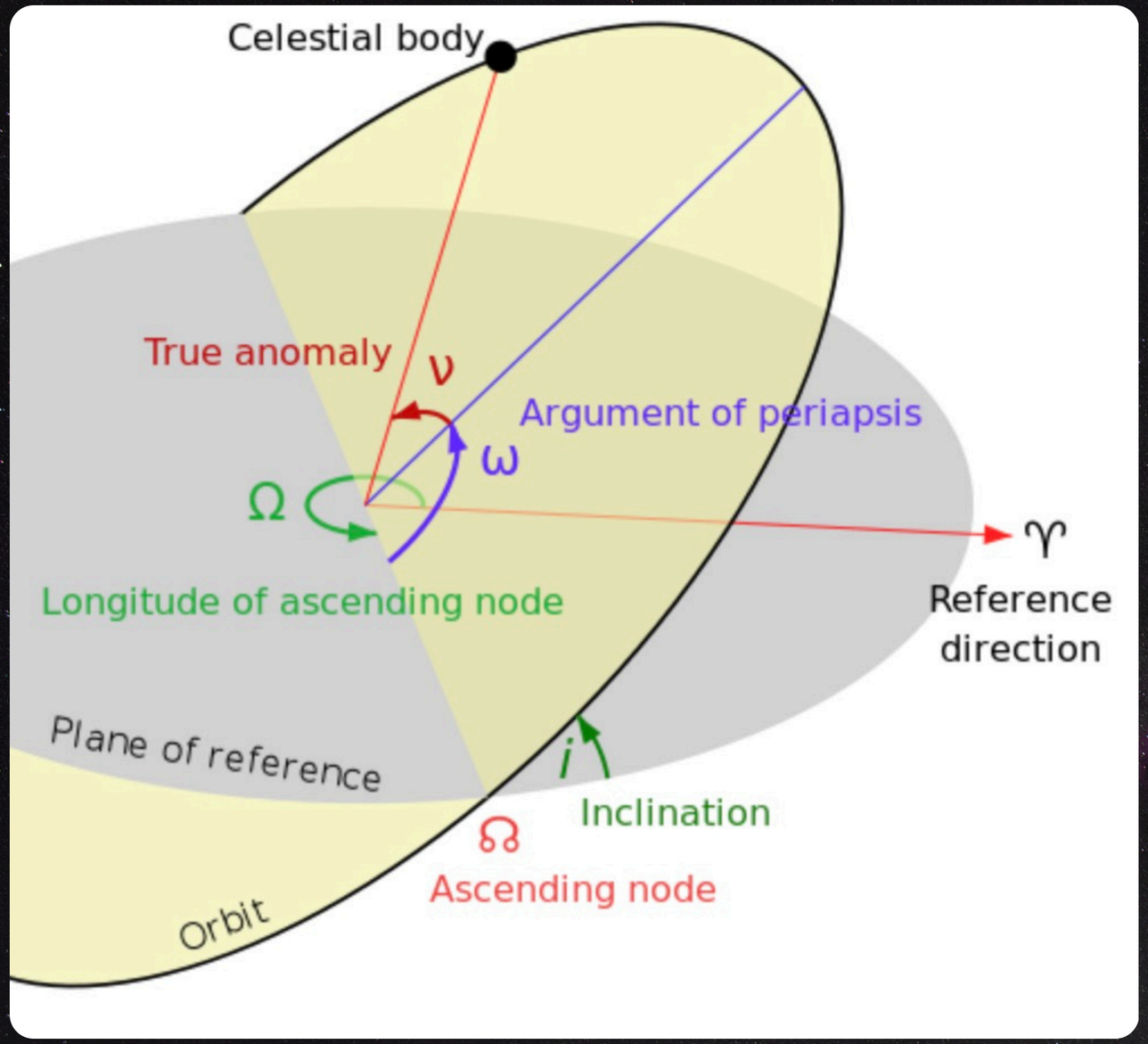
- ➜ Computational cost for SZ calculation: $N \times k \times \frac{n \times P_p}{\Delta t} \sim 50 \times 10^9 \text{ integration steps}$
- ➜ Limited resolution in k
- ➜ Simplified radiative model

Outlook

- ➜ Use parallel computing
- ➜ Explore inclined and retrograde orbits, including Kozaï-Lidov effects
- ➜ Include a more realistic atmospheric model
- ➜ Expand parameter-space exploration
- ➜ Test different planet profiles, such as Venus-like or super-Earth-like planets

THANK YOU!


RELAX ON
KEPLER-16b
THE LAND OF TWO SUNS
WHERE YOUR SHADOW ALWAYS HAS COMPANY


Celestial body
True anomaly
ν
Argument of periapsis
ω
Ω
♈
Longitude of ascending node
Reference direction
Plane of reference
i
Inclination
☊
Ascending node
Orbit

To estimate a realistic surface temperature, it is necessary to account for the warming effect of the planetary atmosphere. We first define the effective equilibrium temperature of the Earth ($T_{\text{eff},\oplus}$), which represents the theoretical temperature of the planet without an atmosphere. According to Equation 10 (with $f_{gh} = 1$), we have:

$$T_{\text{eff},\oplus} = \left[\frac{1-\alpha}{16\pi\sigma}\left(\frac{L_\odot}{d_{E-S}^2}\right)\right]^{1/4} \approx 255 \text{ K} \tag{B.1}$$

where d_{E-S} is the distance between the Earth and the Sun.

However, the actual mean surface temperature of the Earth is measured to be $T_{\text{surf},\oplus} = 288$ K (Kasting et al. 1993). This difference of 33 K is attributed to the natural greenhouse effect of our atmosphere. By comparing the observed value with the theoretical one, we determine the following empirical correction factor:

$$f_{gh} = \frac{T_{\text{surf},\oplus}}{T_{\text{eff},\oplus}} \approx 1.13 \tag{B.2}$$

In our numerical model, this factor is applied to the equilibrium temperature. It provides an approximation of the surface temperature for Earth-like planets.